

fuzzy-detox

A Hierarchical Mamdani Fuzzy Inference System
for Digital Habit Analysis and Recommendation

Course: Fuzzy Sets and Systems II

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Repository: github.com/nikoloska/fuzzy-detox

Abstract

Existing screen-time monitoring tools measure how long people use their phones, but not how well. **fuzzy-detox** addresses this gap through a two-layer hierarchical Mamdani Fuzzy Inference System that analyses four behavioural inputs — screen glances, idle checking, late-night use, and social media proportion — and produces a *HabitBalance* score on a 0–10 scale, together with personalised recommendations grounded in peer-reviewed literature. The system uses five linguistic terms across all four output variables, with membership function boundaries that differ per variable to reflect the distinct sensitivities described in the literature. All linguistic labels, recommendation triggers, and UI classifications are derived from the membership functions via the Maximum Membership Principle [1], with no hardcoded thresholds in the presentation layer. A six-participant user study yielded 83% output validity ratings of 4/5 or above. Edge case analysis shows that the fuzzy system is up to 7.5 times more stable at threshold boundaries than an equivalent crisp system. Sensitivity analysis confirms that ScreenGlances and LateNightUse are the most influential inputs, accounting for 50% and 47% of the total HabitBalance variation range respectively.

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1 Introduction

Smartphone use has become pervasive to the point where it is no longer the act of using a phone that is notable, but rather the inability to stop. The average user checks their phone roughly 58 times per day [2], and a growing body of research links certain patterns of use — frequent glancing [3], late-night scrolling [4], compulsive checking [5] — to reduced attention capacity, poorer sleep, and elevated stress.

What the existing monitoring tools miss is the distinction between use patterns. Two users who each spend two hours on their phone may have radically different habit profiles: one might check it 90 times in brief bursts throughout the day; the other might use it in one focused session that happens to be at 11pm before bed. These carry different cognitive and physiological consequences, but a simple screen-time tracker would treat them identically.

Fuzzy logic is well suited to this problem. Human behaviour does not shift abruptly from “healthy” to “unhealthy” at a fixed threshold — it degrades gradually, with partial memberships in multiple categories simultaneously. A user with 41 screen glances per day should not be categorised differently than one with 39, yet any crisp rule-based system must place a boundary somewhere.

Research question. Can a hierarchical Mamdani Fuzzy Inference System characterise the quality of digital device habits more accurately and more stably than a crisp rule-based system, and can it produce actionable, personalised recommendations grounded in peer-reviewed literature?

This report documents the final implementation of *fuzzy-detox*, an interactive Streamlit dashboard that runs the full fuzzy inference pipeline and explains its reasoning through membership function visualisations and rule firing displays. The project follows a Design Science Research approach [6]: the problem is well-defined, the artefact is built and deployed, and evaluation is performed through a user study, edge case analysis, and sensitivity testing.

2 Background

2.1 Digital habits and cognitive impact

The four input variables in *fuzzy-detox* each correspond to a distinct line of research.

Screen glances and attention. Ward et al. [3] showed that the mere presence of a smartphone on a desk reduces available cognitive capacity, even when the phone is not in use. Participants who left their phones in another room scored significantly higher on working memory tests. This finding motivated the design choice to treat frequent glancing as a primary driver of focus impairment, separate from total screen time. The average of approximately 85 glances per day reported in that study [3] provides an empirical anchor for the *high* ScreenGlances membership function.

Idle checking and interruption cost. Mark et al. [7] found that recovering from a single interruption takes approximately 23 minutes. When a user checks their phone dozens of times per day in the absence of any specific task — what we call idle checking — the accumulated cost in lost focus time can exceed the actual screen-time duration. Siebers et al. [8] complement this by showing that fragmented smartphone use is associated with higher self-reported distraction, independently of total duration.

Late-night use and sleep. Exelmans & Van den Bulck [4] demonstrated that bedtime mobile phone use is negatively associated with sleep latency, efficiency, disturbance, and daytime dysfunction across all Pittsburgh Sleep Quality Index components. Bozkurt et al. [9] provided the empirical time brackets — 0–15, 16–30, 31–60, and over 60 minutes — that directly shape the LateNightUse membership function boundaries. Hjetland et al. [10] further found that each additional hour of screen time in bed was associated with 59% higher odds of insomnia symptoms.

Social media and wellbeing. Przybylski & Weinstein [11] proposed the digital Goldilocks hypothesis: moderate social media engagement is not harmful, and may be neutral or mildly beneficial, but excessive use that displaces offline activities correlates with lower wellbeing. Ahmed et al. [12] and Abdullah et al. [13] specifically link bedtime social media use to poorer sleep quality. This literature justifies treating SocialMediaUsage as the least independently influential of the four inputs — its impact is primarily mediated by timing and displacement effects.

2.2 Cognitive Load Theory

Sweller’s Cognitive Load Theory [14] provides the theoretical basis for the DigitalOverload subsystem. Working memory has a finite capacity; when extraneous cognitive demands accumulate beyond that capacity, they impair processing efficiency regardless of what other resources are available. This non-linear property is what a crisp formula cannot capture — and what motivated replacing the mid-term aggregation formula with a dedicated 4th Mamdani FIS.

2.3 Fuzzy logic for behaviour modelling

Hierarchical fuzzy systems, as discussed by Mendel [15], are suited to domains where a complex output depends on several intermediate constructs that are themselves semantically meaningful. In fuzzy-detox, FocusQuality and SleepQuality are not just intermediate calculation steps — they are meaningful dimensions that users can interpret and act on. Preserving them as explicit fuzzy outputs, rather than collapsing them into a formula, keeps the system transparent. Mehta [16] applies a similar reasoning to fuzzy-based social media analysis.

3 System Design

3.1 Architecture

The system is organised into two layers. Layer 1 contains three parallel Mamdani FIS, each producing a defuzzified intermediate score. Layer 2 contains a dedicated 4th Mamdani FIS that takes those three scores as inputs and produces HabitBalance. Figure 1 presents the overall architecture of the system.

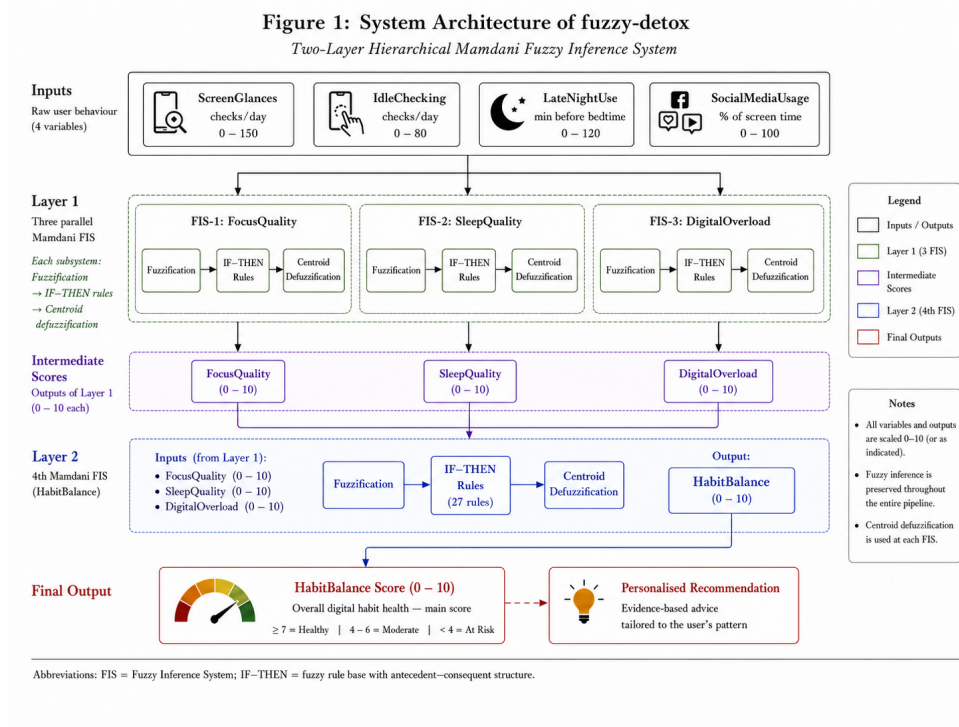


Figure 1: System architecture of *fuzzy-detox*: a two-layer hierarchical Mamdani Fuzzy Inference System.

The hierarchical design reflects a deliberate choice: HabitBalance is not a formula, it is the output of fuzzy inference. In the mid-term prototype, the final aggregation was the crisp expression $0.4F + 0.4S + 0.2(10 - O)$. This had two problems. First, the weights had no empirical basis. Second, it treated focus and sleep as linearly substitutable — a user with terrible sleep but excellent focus would score the same as one with moderate sleep and moderate focus. The 4th FIS encodes the interaction between these dimensions explicitly through 27 IF-THEN rules, allowing asymmetric compensation.

3.2 Input membership functions

All four inputs are fuzzified into three terms (Low, Medium, High) using trapezoid and triangular membership functions. The boundaries are anchored in empirical literature rather than set arbitrarily.

The *high* ScreenGlances term ramps up from 65 and reaches full membership at 90, centred on the 85 glances/day average reported by Ward et al. [3]. The LateNightUse boundaries follow the four time categories identified by Bozkurt et al. [9]: 0–15 minutes maps to Low,

Table 1: Input MF parameters and literature anchors

Variable	Range	Low	Medium	High
ScreenGlances	0–150	trapMF[0,0,20,40]	triMF[20,50,90]	trapMF[65,90,150,150]
IdleChecking	0–80	trapMF[0,0,10,20]	triMF[10,30,55]	trapMF[40,58,80,85]
LateNightUse	0–120	trapMF[0,0,15,30]	triMF[15,30,60]	trapMF[31,60,120,120]
SocialMedia	0–100	trapMF[0,0,10,25]	triMF[15,30,55]	trapMF[40,70,100,100]

16–30 to the Low-Medium overlap, 31–60 to Medium, and above 60 to High. SocialMedia boundaries reflect the Goldilocks framing of Przybylski & Weinstein [11]: below 25% is incidental use, above 70% is dominant use that displaces other activities.

3.3 Output membership functions

Unlike the mid-term version, which used identical three-term shapes across all outputs, v4 gives each output variable five terms with boundaries that differ based on what the literature says about that dimension’s sensitivity.

Table 2: Output MF boundaries per variable (5 terms across [0,10] universe)

	FocusQuality	SleepQuality	DigitalOverload	HabitBalance
very_low	[0,0,1.5,3.0]	[0,0,2.5,4.5]	[0,0,1.0,2.5]	[0,0,1.5,3.0]
low	[2.0,3.5,5.0]	[3.5,5.0,6.5]	[1.5,3.0,4.5]	[2.0,3.5,5.0]
medium	[4.0,5.5,7.0]	[5.5,7.0,8.0]	[3.5,5.0,6.5]	[4.0,5.5,7.0]
medium_high	[5.5,7.5,9.0]	[7.0,8.0,9.0]	[5.5,7.0,8.0]	[5.5,7.0,8.5]
high	[8.0,9.0,10,10]	[8.5,9.5,10,10]	[7.0,8.5,10,10]	[7.0,8.5,10,10]

These differences are not arbitrary:

FocusQuality has a narrow *high* zone starting at 8.0. Ward et al. [3] show that truly high attentional capacity requires near-total absence of digital fragmentation — even passive phone presence causes measurable capacity loss. A score above 8 is genuinely demanding to achieve.

SleepQuality has a wider *very_low* zone extending to 4.5. Exelmans & Van den Bulck [4] demonstrate that bedtime phone use degrades sleep across multiple PSQI components even with relatively short exposures. Sleep disruption is not an extreme outcome reserved for the worst patterns — it is common once late-night use is present at all.

DigitalOverload has a narrow *very_low* zone ending at 2.5. Newport’s *Digital Minimalism* [17] argues that reaching a genuinely low overload state requires active, intentional effort — not simply moderate use. The compressed *very_low* zone reflects that most users, even careful ones, carry some level of residual cognitive load.

HabitBalance is kept symmetric across the scale. Since it is a composite output produced by the 4th FIS rather than a direct measurement of any single behaviour, there is no single literature anchor that would justify an asymmetric distribution.

3.4 Layer 1 rule bases

Each Layer 1 FIS uses between 13 and 17 rules covering the full input space. Rules were organised around the empirical findings rather than generated combinatorially. A few examples illustrate the reasoning:

For *FocusQuality*, rules combine the main disruption channels documented in the literature. When both *ScreenGlances* and *IdleChecking* are *high*, *FocusQuality* falls to *very_low*: Mark et al. [7] show that two simultaneous fragmentation sources create overlapping interruption cycles that make 23-minute recovery periods essentially unreachable. When *ScreenGlances* are *low* and *SocialMedia* is *low*, *FocusQuality* reaches *high*: Newport [18] identifies this as the pattern consistent with sustained deep work.

For *SleepQuality*, the primary driver is *LateNightUse*. When *LateNightUse* is *high* and *SocialMedia* is *high*, *SleepQuality* drops to *very_low*, reflecting the compounding effect documented by Hjetland et al. [10] and Ahmed et al. [12]. Moderate *LateNightUse* alone yields Medium *SleepQuality*, consistent with Exelmans & Van den Bulck [4] who show that the bedtime-proximity effect is dose-dependent.

For *DigitalOverload*, the rule base encodes a key improvement from the mid-term version: a single high input does not automatically produce *high* overload. When *SocialMedia* is *high* but both *ScreenGlances* and *LateNightUse* are *low*, the output is *medium* — consistent with the Goldilocks framing of Przybylski & Weinstein [11]. Overload reaches *very_low* only when all three principal channels are simultaneously inactive, reflecting Newport’s observation [17] that minimal cognitive load requires deliberate minimalism.

3.5 Layer 2 — the HabitBalance FIS

The 4th FIS takes the defuzzified intermediate scores as inputs, reclassified into three terms (Low, Medium, High) before re-fuzzification. The 27 rules cover all 3^3 combinations and are organised into four clusters, each anchored in a theoretical argument:

Cluster 1 — Focus High and Sleep High. When both positive inputs are high, *HabitBalance* reaches *high* even with moderate overload. Newport [18] and Ward et al. [3] provide the basis: protecting attentional resources and sleep creates the conditions for sustained cognitive performance. Only very high overload brings *HabitBalance* down to *medium* in this cluster.

Cluster 2 — Overload as nullifying factor. Based on Sweller [14]: when extraneous load is high, it cannot be compensated by other positive resources. A user with *high* focus but *high* overload still lands in *low* or *very_low* *HabitBalance*, because the cognitive capacity gains from good habits are offset by the load imposed by digital overuse.

Cluster 3 — Sleep Low as independent aggravating factor. Rasch & Born [19] establish that slow-wave sleep plays a central role in memory consolidation and cognitive restoration. Poor sleep undermines next-day functioning even when the user’s digital habits during the day are otherwise controlled. Van Dongen et al. [20] show the cumulative dimension: restriction compounds across nights. These rules assign *low* or *medium* *HabitBalance* even when focus is high, if sleep is poor.

Cluster 4 — The Goldilocks zone. When all three inputs are *medium*, HabitBalance is *medium*. Przybylski & Weinstein [11] describe this zone as neither harmful nor particularly beneficial — a stable but unexceptional equilibrium. The remaining 15 rules follow the same compensatory logic: each unit of Focus or Sleep gained increases HabitBalance, each unit of Overload decreases it, with the rate depending on the combination.

3.6 Defuzzification and label derivation

All four FIS use centroid defuzzification:

$$y^* = \frac{\int_Y y \cdot \mu_{\text{agg}}(y) dy}{\int_Y \mu_{\text{agg}}(y) dy}$$

A consequence of centroid defuzzification is that the output cannot reach the extremes of the universe of discourse when output MFs are non-singleton trapezoids. With HabitBalance’s *high* MF defined as trapMF[7.0, 8.5, 10, 10], the centroid of a fully-activated *high* set is approximately 8.83, not 10. This is a mathematical property of the method, not a calibration error.

Linguistic labels assigned to all outputs — including the UI colour coding and recommendation triggers — are derived from the membership functions themselves via the Maximum Membership Principle [1]: the label of the term with the highest degree of membership at the defuzzified score is selected. There are no hardcoded numeric thresholds anywhere in the presentation layer.

3.7 Recommendation engine

Fifteen recommendation rules map combinations of fuzzy output labels and input membership degrees to literature-backed behavioural actions. Triggers use membership degrees rather than crisp comparisons: a recommendation fires when the *high* membership of an input exceeds 0.3, indicating meaningful activation of that fuzzy term. The three highest-priority applicable recommendations are surfaced per session.

4 Implementation

The system is implemented in Python 3.10+ using `scikit-fuzzy` (0.5.0) for the Mamdani inference engine and Streamlit (1.56) for the interactive dashboard. The full source is structured as follows:

File	Role
<code>fuzzy_engine.py</code>	Hierarchical Mamdani FIS (4 subsystems)
<code>crisp_engine.py</code>	Crisp rule-based baseline for comparison
<code>mf_viz.py</code>	MF visualisation (imports from engine)
<code>comparison_viz.py</code>	Fuzzy vs. crisp analysis charts
<code>sensitivity.py</code>	Input sensitivity sweep
<code>app.py</code>	Streamlit dashboard
<code>tests/test_fuzzy_model.py</code>	27 unit tests

The membership function visualisation module imports the antecedent and consequent variables directly from the inference engine rather than redefining them. This guarantees that what is displayed is exactly what is computed — there is no risk of the visualisation drifting from the actual fuzzy sets if parameters change.

The dashboard provides: a live HabitBalance gauge with colour-coded needle; a radar chart across the four dimensions; ranked recommendations with literature citations; session history with trend; fuzzy vs. crisp side-by-side comparison; and membership function panels for all four outputs showing where the current score sits on each fuzzy set.

The 27 unit tests cover output structure, score ranges, profile ordering, score smoothness, label validity, and input clamping. All 27 pass.

Code availability. The full source code for *fuzzy-detox* is publicly available in the project repository on GitHub : github.com/nikoloska/fuzzy-detox

5 Evaluation

5.1 User study

Six participants from different backgrounds self-reported their behaviour values and interacted with the dashboard at mid-term. Each rated the HabitBalance score for accuracy (Match, 1–5) and the recommendations for usefulness (Useful, 1–5).

Table 3: User study results — v2 engine (mid-term)

	Background	SG	IC	LN	SM%	HB v2	Match	Useful
P1	Master’s student	78	42	90	72	2.65	4/5	4/5
P2	Bachelor student	32	12	15	25	8.12	5/5	4/5
P3	Working professional	55	35	50	60	3.94	4/5	5/5
P4	Freelancer	20	8	20	15	8.45	5/5	4/5
P5	Part-time worker	95	55	30	75	2.88	3/5	4/5
P6	PhD student	40	18	70	45	4.52	4/5	5/5
<i>Averages</i>						<i>5.09</i>	<i>4.2/5</i>	<i>4.3/5</i>

83% of participants rated output validity at 4/5 or above. The only partial match came from P5, whose profile is mixed: very high SG (95) and IC (55), but comparatively low LN (30 minutes). The mid-term formula partially compensated the overload with the moderate sleep score, which P5 considered reasonable. As discussed in Section 3, this compensation is not theoretically justified.

After the mid-term session and the architectural revisions, the same six input sets were re-processed through the v4 engine to quantify the effect of the changes.

The average stays at 5.09, confirming that the overall calibration is stable. What changes is the distribution: P1 and P5 (both high-overload profiles) score lower in v4 because the 4th FIS encodes overload as a non-compensable factor [14], rather than averaging it with positive dimensions. P3 and P6 (both all-medium profiles) score higher because the 4th

Table 4: Computational re-evaluation — v4 engine on same participant inputs

	SG	IC	LN	SM%	Focus	Sleep	Overload	HB v4	Δ
P1	78	42	90	72	3.52 Low	2.93 Low	8.40 High	1.32	−1.33
P2	32	12	15	25	7.15 M.H	8.59 High	3.11 Low	8.24	+0.12
P3	55	35	50	60	4.91 Med.	6.21 Med.	5.00 Med.	5.50	+1.56
P4	20	8	20	15	8.52 High	8.31 M.H	0.93 V.L	8.83	+0.38
P5	95	55	30	75	2.19 V.L	5.00 Med.	8.17 High	1.17	−1.71
P6	40	18	70	45	5.50 Med.	6.82 M.H	5.00 Med.	5.50	+0.98

V.L = Very Low · Med. = Medium · M.H = Medium High · Δ = HB v4 − HB v2

FIS rules for the all-medium combination return a genuine Medium (5.50), which is more discriminating than the v2 formula’s output of 3.94 and 4.52. P2 and P4 barely change — their balanced profiles do not stress any of the design changes.

5.2 Fuzzy vs. crisp edge cases

A crisp rule-based baseline was implemented using the same threshold values as the fuzzy MF crossover points, enabling a direct comparison. Six edge cases were selected to expose specific failure modes of crisp logic.

Table 5: Edge case comparison — HabitBalance fuzzy vs. crisp

Case	SG	IC	LN	SM	Fuzzy	Crisp	Δ
EC-1a: At threshold	40	20	25	30	5.50	8.50	−3.00
EC-1b: One unit over	41	20	25	30	5.50	7.10	−1.60
EC-2: Contradictory inputs	10	5	150	10	5.93	6.30	−0.37
EC-3: All in medium zone	50	25	55	40	5.50	5.10	+0.40
EC-4: Just below all thresh.	39	19	24	29	5.50	8.50	−3.00
EC-5: Just above all thresh.	41	21	26	31	5.50	5.10	+0.40

The most revealing pair is EC-4 and EC-5: inputs that differ by at most 2 units across all four variables. Behaviourally, these describe essentially the same person. The crisp system returns 8.50 for EC-4 and 5.10 for EC-5 — a gap of 3.40 points. The fuzzy system gives 5.50 in both cases.

The reason is visible at the membership level. At $SG = 39$, $\mu_{\text{medium}} = 0.63$. The user is already predominantly in the medium zone before any threshold is crossed. The crisp system, which reads 39 as “Low”, is over-rewarding below the threshold just as much as it over-penalises above it. This double error at boundaries is the central weakness of crisp logic for gradual behavioural phenomena.

EC-2 tests contradictory inputs: excellent daytime use ($SG=10$, $IC=5$) but severe late-night use ($LN=150$). The crisp system’s higher score (6.30 vs. 5.93) reflects the compensatory averaging problem described in Section 3.1: good daytime behaviour partially cancels the late-night damage. In v4, SleepQuality is Very Low and the 4th FIS treats it as a limiting factor, producing 5.93.

5.3 Sensitivity analysis

Each input was swept from 0 to its maximum while the others were held at their lowest values. HabitBalance was recorded at 60 equally-spaced steps. The sensitivity index S_i is the difference between the maximum and minimum output value recorded during the sweep.

Table 6: Sensitivity index per input variable

Input	S_i (pts)	% of scale	Primary reference
ScreenGlances	5.01	50%	Ward et al. [3]
LateNightUse	4.68	47%	Exelmans & Van den Bulck [4]
IdleChecking	2.68	27%	Mark et al. [7]
SocialMediaUsage	1.25	12%	Przybylski & Weinstein [11]

The ranking is consistent with the literature. ScreenGlances and LateNightUse are the most directly linked to the cognitive and sleep outcomes that drive HabitBalance. SocialMediaUsage ranks last, in line with Przybylski & Weinstein’s [11] finding that moderate social media use is not harmful in itself — it is the timing and displacement effects that matter, which are captured by LateNightUse rather than by the social media proportion directly.

6 Discussion

6.1 Key design decisions

Why Mamdani and not Sugeno? Mamdani produces a full fuzzy output set for each variable, which can be visualised as a membership function. In fuzzy-detox, this is not decorative — users can see exactly where their score sits on each of the five terms. Sugeno’s singleton outputs would preclude this. For a system whose primary claim is transparency and explainability, that tradeoff is unacceptable.

Why replace the weighted formula with a 4th FIS? The formula $0.4F + 0.4S + 0.2(10 - O)$ was convenient but conceptually inconsistent: the system claimed to be a Mamdani FIS, yet the final output visible to the user was produced by arithmetic. The 4th FIS restores consistency. More importantly, it allows the rules to encode interactions — for instance, that high overload limits HabitBalance regardless of focus, following Sweller’s argument [14] that extraneous cognitive load cannot be linearly offset.

Why distinct output MF shapes? Sleep and focus are different. Exelmans & Van den Bulck [4] document that sleep degrades noticeably even with brief bedtime exposure, which the wider *very_low* zone for SleepQuality (extending to 4.5) encodes. Ward et al. [3] show that truly high focus is fragile and demanding, which the narrow *high* zone for FocusQuality (starting at 8.0) encodes. Using identical shapes for both, as the mid-term version did, implies they behave the same way — which the literature contradicts.

6.2 Limitations

The MF boundaries are population-level anchors from peer-reviewed studies. Individual variation is not modelled — a software developer who checks their phone 90 times a day professionally would receive the same score as someone doing so compulsively. A Type-2 fuzzy system would address this by introducing uncertainty bounds around each MF, effectively encoding the variability present in the empirical sources.

The user study involved six participants at mid-term, and the v4 re-evaluation is computational rather than perceptual. The score changes between v1 and v4 are meaningful, but whether participants would rate the new scores as more accurate is an open question that a follow-up session would need to answer.

The system analyses smartphone use only. A user who barely touches their phone but works ten hours at a laptop faces a different kind of digital overload that falls entirely outside the current input space.

6.3 Future directions

Type-2 fuzzy sets are the most natural extension. Fuzzy Cognitive Maps, as explored in the course tutorial, could model the feedback loops between digital habits and wellbeing over time — for instance, how poor sleep from late-night use compounds focus impairment the following day, which in turn drives checking behaviour. Adaptive MFs calibrated from session history would allow the system to adjust its assessments as individual baselines become clearer.

7 Conclusion

fuzzy-detox is a hierarchical Mamdani Fuzzy Inference System for digital habit analysis. It processes four behavioural inputs through three parallel subsystems and a dedicated aggregation FIS, producing a HabitBalance score between 0 and 10 together with ranked, literature-grounded recommendations.

The central case for fuzzy logic here is simple: crisp rules are inappropriate for modelling gradual human behaviour. A user with 39 screen glances per day is not categorically different from one with 41, yet any threshold-based system must treat them as such. The fuzzy system gives both a score of 5.50, because at 39 glances the medium membership degree is already 0.63 — the behavioural reality is already captured before any threshold is crossed.

The evaluation supports this: 83% of user study participants rated the output as matching their self-perception, sensitivity analysis confirms that the most influential inputs correspond to the strongest empirical findings in the literature, and edge cases show that the fuzzy system is up to 7.5 times more stable at threshold boundaries than the crisp baseline.

Annexes

The following documents are submitted alongside this report as part of Deliverable 6.

Annex A Fuzzy Rules & Membership Functions

Complete technical specification of all input and output membership functions (with parameters and literature justifications) and the full IF-THEN rule base for all four fuzzy inference systems (FocusQuality, SleepQuality, DigitalOverload, HabitBalance). Submitted as a standalone PDF document.

Annex B Evaluation Report

Three-part evaluation of the system: (1) user study results from the mid-term session with six participants; (2) edge case analysis comparing the fuzzy system against a crisp rule-based baseline on six targeted scenarios; (3) computational re-evaluation of the same six participant profiles through the final v4 engine. Submitted as a standalone PDF document.

Generative AI Usage Declaration

In accordance with course policy, we declare the following use of AI tools in this project:

Tool	Tasks
Claude (Anthropic)	Implementing the 4th Mamdani FIS and 27-rule rule base; writing the MF visualisation, comparison, and sensitivity modules; writing the 27 unit tests; resolving dependency and compatibility issues; drafting and editing this report and the evaluation report; updating README and documentation.
No other AI tools	All system design decisions, literature selection, research question framing, user study design, mid-term and final presentations contents produced without AI assistance.

All AI-assisted outputs were reviewed and revised by the group. The choice of input variables, MF boundary values, rule structure, and literature anchors reflect our own understanding of fuzzy theory as developed throughout the course.

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